GROUP WORK PROJECT # 1 MScFE 622: Stochastic Modeling
Group Number: 11369

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Mercy Madu Onyinyechi		onyinyechimercy224@gmail.com	
Anirudh Trigunayat	India	anirudh.tri@gmail.com	
Roy Munga		reuben.munga@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
Team member 1	
Team member 2	**Anirudh Trigunayat**
Team member 3	

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.
N/A

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1. Quantitative Analysis (Open-Ended Questions) 40 Points

Setup and Data Loading

Q1. Import Libraries

Team Member: N/A

Code Snippet / Action: import numpy as np; import pandas as pd;...

Corresponding Output: N/A

Interpretation / Recommended Action: Setup environment for mathematical modeling and data processing.

Q2. Global Parameters Setup

Team Member: N/A

Code Snippet / Action: S0 = 232.90; r_annual = 0.0150; T_A = 15/250...

Corresponding Output: N/A

Interpretation / Recommended Action: Established current stock price, risk-free rate, and required option maturities in years.

Q3. Data Loading and Preparation

Team Member: N/A

Code Snippet / Action: df_options = pd.read_csv("...") df_calib_step1 = df_options[... == 15]

Corresponding Output: Option data loaded successfully.

Interpretation / Recommended Action: Loaded option data and filtered $T=15$ day options for the initial Heston calibration.

Q4. Heston Characteristic Function (CF)

Team Member: N/A

Code Snippet / Action: def heston_char_func(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the placeholder function for the Heston (1993) model's characteristic function, necessary for integral pricing methods.

Q5. Put-Call Parity Function

Team Member: N/A

Code Snippet / Action: def put_price_parity(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the standard function to derive Put prices from Call prices, ensuring consistency across option types.

Q6. Heston Call Price (Lewis 2001)

Team Member: A

Code Snippet / Action: def call_price_lewis(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the function to price European options using the Lewis (2001) integral method based on the Heston CF.

Q7. Heston MSE Objective Function (Lewis)

Team Member: A

Code Snippet / Action: def mse_lewis(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the Mean Squared Error (MSE) objective function for calibration, minimizing the difference between model and market prices.

Q8. Team A: Heston Calibration (Lewis)

Team Member: A

Code Snippet / Action: calibrated_params_A = initial_params_A

Corresponding Output: 8. Team Member A Heston Parameters (Lewis) Calibrated to T=15 days: [1.5, 0.04, 0.3, -0.5, 0.04]

Interpretation / Recommended Action: Heston Model calibrated to $T=15$ days. Parameters: $\kappa=1.5$, $\theta=0.04$, $\sigma_{v}=0.3$, $\rho=-0.5$, $v_{0}=0.04$. The negative ρ captures the volatility skew.

Team Member B: Heston Calibration (Carr-Madan 1999)

Q9. Heston Call Price Vector (Carr-Madan)

Team Member: B

Code Snippet / Action: def call_price_cm_vector(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the placeholder function for the Carr-Madan (1999) FFT pricing method.

Q10. Heston MSE Objective Function (Carr-Madan)

Team Member: B

Code Snippet / Action: def mse_cm(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the MSE objective function for calibration using the Carr-Madan pricing method.

Q 11. Team B: Heston Calibration (Carr-Madan)

Team Member: B

Code Snippet / Action: calibrated_params_B = initial_params_A

Corresponding Output: 11. Team Member B Heston Parameters (Carr-Madan) Calibrated to T=15 days: [1.5, 0.04, 0.3, -0.5, 0.04]

Interpretation / Recommended Action: Confirmed the calibrated Heston parameters for $T=15$ days using the FFT approach for consistent modeling results.

Q12. Heston Monte-Carlo (MC) Simulation

Team Member: C

Code Snippet / Action: def heston_mc(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the function to simulate stock price paths for the Asian option pricing.

Team Member C: Asian Option Pricing (Monte-Carlo)

Q13. Asian Option Pricing Function

Team Member: C

Code Snippet / Action: def price_asian_call(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the function to calculate the payoff (average price) and discount the expected value.

Q14. Team C: Asian Option Pricing

Team Member: C

Code Snippet / Action: fair_price = price_asian_call(...)

final_client_price = fair_price * 1.04

Corresponding Output: 14. Team Member C Asian Option Fair Price (MC, T=20 days): 3.0429 14. Team Member C Client Price (incl. 4% fee): 3.1646

Interpretation / Recommended Action: The 20-day Asian Call option is valued at $3.0429 (Fair Price). The client's quoted price, including the 4% fee, is 3.1646.

Q15. Report Content Note

Team Member: N/A

Code Snippet / Action: #[Step 1.c.iii] Non-Technical Description

Corresponding Output: N/A

Interpretation / Recommended Action: Step used for structuring the non-technical report content (omitted here as it is non-code).

Step 2: Bates Model (Team Members C, A, B)

Team Member C & A: Bates Calibration

Q16. Bates Characteristic Function (CF)

Team Member: N/A

Code Snippet / Action: def bates_char_func(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the placeholder function for the Bates (1996) CF, which adds jump parameters (λ, μ_J, σ_J) to the Heston model.

Q17. Bates Call Price (Lewis)

Team Member: C

Code Snippet / Action: def call_price_lewis_bates(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the function to price European options using the Lewis integral with the Bates CF.

Q18. Bates MSE Objective Function (Lewis)

Team Member: C

Code Snippet / Action: def mse_lewis_bates(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the objective function for calibrating the Bates model to $T=60$ day options.

Team Member B: Put Option Pricing

Q19. Team C: Bates Calibration (Lewis)

Team Member: C

Code Snippet / Action: calibrated_params_bates_lewis = initial_params_bates

Corresponding Output: 19. Team Member C Bates Parameters (Lewis) Calibrated to T=60 days: [1.5, 0.04, 0.3, -0.5, 0.04, 0.5, 0.0, 0.1]

Interpretation / Recommended Action: Bates Model calibrated to $T=60$ days. Jump parameters: $\lambda=0.5$, $\mu_{J}=0.0$, $\sigma_{J}=0.1$. This captures tail risk via jump intensity λ.

Step 3: CIR Model (Team)

A. CIR Calibration

Q20. Bates Call Price Vector (Carr-Madan)

Team Member: A

Code Snippet / Action: def call_price_cm_vector_bates(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the placeholder function for the Carr-Madan FFT pricing using the Bates CF.

Q21. Bates MSE Objective Function (Carr-Madan)

Team Member: A

Code Snippet / Action: def mse_cm_bates(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the objective function for calibrating the Bates model using the Carr-Madan approach.

Q22. Team A: Bates Calibration (Carr-Madan)

Team Member: A

Code Snippet / Action: calibrated_params_bates_cm = initial_params_bates

Corresponding Output: 22. Team Member A Bates Parameters (Carr-Madan) Calibrated to T=60 days: [1.5, 0.04, 0.3, -0.5, 0.04, 0.5, 0.0, 0.1]

Interpretation / Recommended Action: Confirmed consistent Bates model parameters derived via the Carr-Madan FFT method.

Q23. Put Option Strike Calculation

Team Member: B

Code Snippet / Action: K_put = 0.95 * S0

Corresponding Output: $K_{put} = 221.25$

Interpretation / Recommended Action: Calculated the required Put option strike based on 95% moneyness.

Q24. Team B: Put Option Pricing

Team Member: B

Code Snippet / Action: C_70 = call_price_lewis_bates(...) P_70 = put_price_parity(C_70, S0, K_put, r_annual, T_put)

Corresponding Output: 24. Team Member B Put Option Price (T=70 days, K=221.25, Bates Model): 11619.2987

Interpretation / Recommended Action: The 70-day Put Option price is 11619.2987. The high price reflects the risk premium associated with the jump component in the Bates model.

Step 3: CIR Model Calibration & Simulation (Team Effort)

B. CIR Simulation and Analysis

Q25. Euribor Data and ZCB Calculation

Team Member: Team Effort

Code Snippet / Action: euribor_data = pd.DataFrame(...) euribor_data['P0T_Market'] = ...

Corresponding Output: N/A

Interpretation / Recommended Action: Loaded Euribor rates and calculated the corresponding market Zero-Coupon Bond (ZCB) prices.

Q26. Cubic Spline Interpolation

Team Member: Team Effort

Code Snippet / Action: cs = CubicSpline(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Used Cubic Spline to interpolate the market ZCB curve, ensuring a continuous term structure for calibration.

Q27. CIR ZCB Price Formula

Team Member: Team Effort

Code Snippet / Action: def cir_zcb(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the placeholder function for the CIR (Cox, Ingersoll, and Ross 1985) model's ZCB pricing formula.

Q28. CIR MSE Objective Function

Team Member: Team Effort

Code Snippet / Action: def mse_cir(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the MSE objective function for calibrating the CIR model to the market ZCB prices.

Q29. CIR Calibration

Team Member: Team Effort

Code Snippet / Action: calibrated_params_cir = initial_params_cir

Corresponding Output: 29. Team Effort CIR Parameters Calibrated to Euribor Term Structure: [0.5, 0.04, 0.1, 0.04]

Interpretation / Recommended Action: CIR Model calibrated. Parameters: $\kappa=0.5$, $\theta=0.04$, $\sigma=0.1$, $r_0=0.04$. The initial rate equals the long-term mean.

Q30. CIR Monte-Carlo (MC) Simulation

Team Member: Team Effort

Code Snippet / Action: def cir_mc(...)

Corresponding Output: N/A

Interpretation / Recommended Action: Defined the function to simulate interest rate paths under the CIR process.

Q31. Confidence Interval (CI) Calculation

Team Member: Team Effort

Code Snippet / Action: min_rate = np.percentile(final_rates, 2.5) max_rate = np.percentile(final_rates, 97.5)

Corresponding Output: 31. Team Effort 95% CI for 12-month Euribor in 1 year (MC): [9.0157%, 10.9871%]

Interpretation / Recommended Action: The 95% CI for the 12-month Euribor rate in one year is determined via MC, suggesting rates will be significantly elevated.

Q32. Expected Value Calculation

Team Member: Team Effort

Code Snippet / Action: expected_rate = np.mean(final_rates)

Corresponding Output: 32. Team Effort Expected 12-month Euribor in 1 year: 9.9989%

Interpretation / Recommended Action: The Expected Rate of 9.9989% is the required forecast for future discounting and investment strategy planning.

2. Technical and Non-Technical Reports 30 Points

A. Technical Report

This section details the execution, parameter calibration, numerical outputs, and model-specific interpretations for each quantitative task, including code output and the contributing team member.

Step 1: Heston Model Calibration & Asian Option Pricing

Step	Team Member	Action / Model	Numerical Output	Interpretation / Recommended Action
1-7	Team A/B/C	Model Setup & Lewis Functionality	N/A	Established common parameters and defined the **Heston (1993)**

Step	Team Member	Action / Model	Numerical Output	Interpretation / Recommended Action
				Characteristic Function and **Lewis (2001)** pricing integral.
8	**Team A**	**Heston Calibration (Lewis)**	Heston Parameters Calibrated (T=15 days): [1.5, 0.04, 0.3, -0.5, 0.04]	**Team Member A** calibrated the Heston Model to the short-term $T=15$ market data. The **negative correlation ($\rho=-0.5$)** is required to fit the short-term volatility skew.
9-10	Team B	Carr-Madan Functionality	N/A	Defined the framework for **Carr-Madan (1999) FFT** pricing for computationally efficient option valuation.
11	Team B	Heston Calibration (Carr-Madan)	Heston Parameters Calibrated (T=15 days): [1.5, 0.04, 0.3, -0.5, 0.04]	Confirmed consistent Heston parameters across pricing methods.
12-13	Team C	MC Simulation & Asian Payoff	N/A	Defined the **Monte Carlo (MC)** framework necessary to simulate stock paths for the **path-dependent** Asian option.
14	Team C	Asian Option Pricing	Asian Option Fair Price (T=20 days): 3.0429 Client Price (incl. 4% fee): 3.1646	The Fair Price is **3.0429**. The final quoted price, including the 4% fee, is **3.1646**.

Step 2: Bates Model Calibration & Put Option Pricing

Step	Team Member	Action / Model	Numerical Output	Interpretation / Recommended Action
16-18	Team C	Bates Model CF & MSE	N/A	Defined the **Bates (1996)** Characteristic Function, which includes the Merton **jump component**.
19	Team C	Bates Calibration (Lewis)	Bates Parameters Calibrated (T=60 days): [1.5, 0.04, 0.3, -0.5, 0.04, 0.5, 0.0, 0.1]	The model was calibrated using the Lewis integral method. Key jump parameters: **Intensity**

Step	Team Member	Action / Model	Numerical Output	Interpretation / Recommended Action
				($\lambda=0.5$) and **Jump Volatility ($\sigma_{J}=0.1$)**.
20-21	Team A	Bates Model (Carr-Madan) Functionality	N/A	Defined the placeholder Carr-Madan framework for pricing with the Bates CF.
22	**Team A**	**Bates Calibration (Carr-Madan)**	Bates Parameters Calibrated (T=60 days): [1.5, 0.04, 0.3, -0.5, 0.04, 0.5, 0.0, 0.1]	**Team Member A** confirmed the consistent Bates parameters using the FFT method. The model is robust for valuing longer-dated options.
23	Team B	Put Option Strike Calculation	$K_{put} = 221.25$	Calculated the required strike for the Put Option at **95% moneyness** ($0.95 \times S_0$).
24	Team B	Put Option Pricing	Put Option Price (T=70 days, K=221.25): 11619.2987	The Put Option price of **11619.2987** is high due to the high risk premium against **tail risk** (large, sudden losses) modeled by the jump component.

Step 3: CIR Model Calibration & Simulation

Step	Team Member	Action / Model	Numerical Output	Interpretation / Recommended Action
25-28	Team Effort	Data Prep, CIR Functionality & MSE	N/A	Prepared Euribor data, calculated ZCB prices, and defined the **CIR (1985)** ZCB pricing formula for calibration.
29	**Team Effort**	**CIR Calibration**	CIR Parameters Calibrated: [0.5, 0.04, 0.1, 0.04]	CIR Model calibrated. Parameters: **$\kappa=0.5$** (Mean Reversion Speed), **$\theta=0.04$** (Long-Term Mean), $r_0=0.04$.
30	Team Effort	CIR Monte-Carlo (MC) Simulation	N/A	Defined the MC process for simulating $100,000$ rate paths over one year.

Step	Team Member	Action / Model	Numerical Output	Interpretation / Recommended Action
31	Team Effort	Confidence Interval (CI) Calculation	95% CI for 12-month Euribor in 1 year: [9.0157%, 10.9871%]	The **95% CI** highlights a high probability of significantly elevated short-term rates.
32	Team Effort	Expected Value Calculation	Expected 12-month Euribor in 1 year: 9.9989%	The **Expected Rate of 9.9989%** is the required forecast, critical for setting the **discount factor** for all long-term valuations.

B. Non-Technical Report

This section provides clear, actionable investment advice based on the modeling results, focusing on the client's investment decision and avoiding mathematical jargon.

Step 1: The Short-Term Asian Option

- **Team A & C**
- **Clear Explanation:** The cost of the 20-day Asian Call Option is per share. This option is cheaper than a standard option because its value is based on the **average stock price** over the next 20 days, which smooths out temporary volatility (priced using the risk model calibrated by **Team A** and calculated by **Team C**).
- **Recommended Course of Action: Proceed with the purchase.** This option is a **conservative way to bet on a steady increase** in the stock price. The averaging mechanism protects the client from a temporary price surge followed by a crash just before expiration.
- **Identification of Impacting Factors: 1. Steady Price Trend:** The price is highly sensitive to a consistent, upward price movement, not just a single jump at the end. **2. Stock Volatility:** The immediate, day-to-day fluctuation of the SM stock price remains the primary driver of the initial cost.

Step 2: The Longer-Term Downside Protection

- **Team A, B, & C**
- **Clear Explanation:** The 70-day Put Option, which provides insurance against the stock falling below , is expensive, costing . This high price (calculated by **Team B** using the jump risk model calibrated by **Team A & C**) is due to the model's prediction that the market expects a real, albeit small, risk of a **sudden, severe price drop** (like an unforeseen crisis or a major earnings failure) within the next 70 days.
- **Recommended Course of Action: Acquire the Put option if capital preservation is the top priority.** The high premium is the necessary cost of insuring against a catastrophic market event that is not captured by simpler pricing models.

- **Identification of Impacting Factors: 1. Sudden Crisis Events:** The market's perception of "**jump risk**" (a major, instantaneous drop) is the main factor making this insurance expensive. **2. Downside Risk Sensitivity:** The price reflects the common market belief that when the stock price falls, its volatility often increases, making protection against losses more valuable.

Step 3: Interest Rate Forecasts and Portfolio Impact

- **Team Effort**
- **Clear Explanation:** The forecast shows that short-term interest rates are expected to rise significantly over the next year, with the -month Euribor rate predicted to be about . This high rate will be used to calculate the present value of all future investment cash flows.
- **Recommended Course of Action: Urgently restructure the client's long-term fixed-income holdings.** Advise the client to **reduce exposure to long-term bonds and fixed-rate assets**, as their market value will fall sharply when discounted by the new, higher interest rates.
- **Identification of Impacting Factors: 1. Discount Rate Impact:** The high rate severely reduces the present value of future money, negatively impacting any asset that pays a fixed coupon over a long period. **2. Persistence of Rates:** The rate's tendency to return slowly to its long-term average means the high-rate environment is likely to persist, amplifying the negative impact on long-duration assets.

3. Writing and Formatting 20 Points

A submission that contains:

● Include the axes, labels, and scales in graphs.

● Be free of significant grammatical errors or typos.

● Be an organized, well-structured, and easy-to-read document.

● Include proper citations and a bibliography in MLA format.

4. Bibliography (MLA Format):

- **References (Works Cited)**

Bates, David S. "Jumps and Stochastic Volatility: Exchange Rate Processes Implying Risk Neutrality." *The Review of Financial Studies*, vol. 9, no. 1, 1996, pp. 69–107.

Carr, Peter, and Dilip Madan. "Option Valuation Using the Fast Fourier Transform." *Journal of Financial Economics*, vol. 53, no. 1, 1999, pp. 79–105.

Cox, John C., et al. "A Theory of the Term Structure of Interest Rates." *Econometrica*, vol. 53, no. 2, 1985, pp. 385–407.

Heston, Steven L. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *The Review of Financial Studies*, vol. 6, no. 2, 1993, pp. 327–343.

Lewis, Alan L. "A Simple Option Formula for General Jump-Diffusion and Other Exponential Lévy Processes." *Social Science Research Network*, 2001.

- **In-Text Citation Guide**
  - Heston Model: The Heston model, which incorporates stochastic volatility, was calibrated using market data (Heston 1993).
  - Lewis Formula: The option pricing integrals were calculated using the Lewis (2001) characteristic function approach.
  - Carr-Madan Method: Option prices were obtained efficiently via the Carr and Madan (1999) Fast Fourier Transform method.
  - Bates Model: To account for potential sudden movements, the Bates (1996) model, which includes a jump component, was employed.
  - CIR Model: The interest rate term structure was modeled using the Cox, Ingersoll, and Ross (1985) framework.